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BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Ethical Analysis Report

Group ID: R26-DS-010

Project Title: Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

Student ID: IT22178640

Individual Component Title: AI-Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution

	Identification and Analysis
Data Privacy Concerns	This component uses the Tracking Individual Performance with Sensors data set collected in 2018, known as TILES-2018, and published as "*TILES-2018, a longitudinal physiologic and behavioral data set of hospital workers*" by Mundnich et al. in Scientific Data (2020). The data were collected by the Signal Analysis and Interpretation Laboratory of the University of Southern California (USC), led by Dr. Shrikanth Narayanan, in partnership with USC Keck Hospital in Los Angeles. The study followed 212 hospital workers for about ten weeks. Wearable sensors recorded their heart rate and activity, and repeated surveys recorded their stress. The consent process is described in the data set paper. All steps of the study were carried out under the approval of the Institutional Review Board of the USC Health Sciences Campus (study ID HS-17-00876). Participants were volunteers who worked at the hospital. They were reached through emails from hospital leaders, employee meetings, flyers and information tables in the cafeteria. After a screening questionnaire, each eligible person was guided through identity verification and informed consent in the study's mobile application. Of 365 eligible people, 212 gave their consent and completed enrolment at an in-person session, where each was assigned a unique participant ID. Eight participants later chose to leave the study. I received the data after signing a data use agreement with USC. The agreement was signed for the university by Ms. Judy Genovese of the USC Stevens Center for Innovation, and I have acknowledged them in my individual thesis. The records I received carry coded participant identifiers, and the demographic fields were curated by the data providers to prevent re-identification. I made no attempt to identify any participant, to obtain a coding key or to contact anyone. The data are used only for internal research on this project, they are shared with no outside party, and only aggregate results are reported. The application displays synthetic caregiver names and ward labels, so a demonstration

	reveals no real person, and the model needs no name, age or gender because it works from heart rate, activity, sleep and survey answers. If this component is later used with care home staff, Sri Lanka's Personal Data Protection Act, No. 9 of 2022, will guide that work. Caregivers will be told what is recorded and will give their consent, and ethics approval will be obtained before any data are collected.
Data Security	The research data were held by the USC team on a secure FTP server. Once the data use agreement was in place, the team gave me access credentials: the host address, port 21, a username and a password. I used these credentials in two ways. I connected with the FileZilla client to browse the data directories and to download files for inspection. I also connected Google Colab to the same server through FTPS, which encrypts the login and the transfer, and the notebooks read and trained on the data from there. The password is typed in when the notebook runs and is never saved in a notebook, and the credentials are not shared with anyone. The downloaded files, the cached tables and the results are kept in a private Google Drive folder. In the web application, sign-in is handled by Supabase Authentication. For every request, the backend validates the user's access token, loads the user's role from the profile table in the Supabase PostgreSQL database and checks that the caregiver being requested falls within that user's scope. The roles are caregiver, supervisor and administrator, so a supervisor sees only the assigned team, and an administrator sees all monitored caregivers. Row-level security is enabled on the sensitive tables, which means the database itself limits the rows that each signed-in user can read. The Supabase service-role key is able to bypass row-level security, so it is stored only on the server as an environment variable, never sent to the browser, and the same scope checks are repeated inside each backend route. An audit log shared across the CareSense modules supports these controls. It records which user viewed or changed a record and at what time, so access to sensitive information can be reviewed afterwards. The model registry adds a second record, since it shows which model version produced each score. Because the application displays synthetic caregiver names and ward labels, the information visible on screen is further limited.
Bias and Fairness	The data come from hospital staff in the United States who volunteered for the study, while the intended users work in care homes in Sri Lanka, where shifts, duties and working conditions differ. The results are therefore reported as evidence from one working population, each with a confidence interval. The stress label is a self-reported rating on a five-point scale, and people may use such a scale differently. The cut-off of three for elevated stress is a working definition for this

	research and carries no clinical meaning. To reduce unfair differences between people, each caregiver's heart rate is compared with that person's own recent pattern, and all testing was carried out on participants who took no part in training.
Transparency and Accountability	Individual and group theses have been written for this work. My individual thesis documents the methodology of this component in full. It covers the data and its source, the allocation of participants, the construction of features at each survey cut-off, the two-stage model, the evaluation measures and the ethical considerations. The group thesis describes how the four CareSense components were combined into one application. All the research work sits in multiple Google Colab notebooks. Each stage starts with a markdown cell that explains what I am doing and the reason for it, so anyone opening the notebook can follow the work from the first cell to the last. Every random step uses seed 42, and I recorded the library versions. I also saved the participant allocation before building any feature, which means anyone can rerun the work and arrive at the same results. I have also prepared a separate handoff document that sets out the methodology in detail. All experiments, including the ablations, are reported with their results. Every experiment is reported with its result. These include the final test score and its confidence interval, the repeated validation by participant, the comparison with the previous stress response alone, the tests with and without proximity centrality, and the experiments that leave out the previous response. Results that gave no improvement are reported as well. The application and the notebooks are maintained in a GitHub repository, so every change can be traced. Within the application, each score is labelled as a model estimate of the probability of elevated stress. Every estimate is shown with SHAP-based factors that explain what influenced it, and the model registry displays which model version produced it. The audit log records user actions, and supervisors review each recommendation and mark it as reviewed.
Impact on Stakeholders	Caregivers are the main group concerned. Caregivers could receive support earlier and share the workload more fairly. Before any real monitoring begins, staff would be informed and asked for their consent. Feedback from a caregiver working in elder care shaped the requirements. The caregiver found the tier colors and the list of factors behind each score easy to follow, and advised that a percentage should be described as a model estimate. Supervisors can see early that a caregiver is under strain and why. That helps them speak with the person in a supportive way and plan the work around it. A representative of Remark Elder Care advised that a

	score should open that conversation and that a suggested substitute should reflect the skill mix, shift pattern and minimum staffing of a unit. These points were recorded as requirements. Residents benefit indirectly, since a rested and supported team is better placed to give safe and good-quality care. The data provider and the volunteers of TILES-2018 are respected through the terms of the data use agreement, the reporting of aggregate results only, and the acknowledgement of the USC research team and the 212 participants in my thesis. The research also supports Sustainable Development Goal 3, Good Health and Well-being, by addressing the sustainability of the caregiver workforce.